

Homework 10 File Submission

Explain using the properties of matrices, either each are reflexive, irreflexive, symmetric, antisymmetric, and/or transitive. For each property, tell why or why not.

$$\mathbf{a)} \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

Reflexive

Irreflexive

Symmetric

Antisymmetric

Transitive

$$\mathbf{b)} \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Reflexive

Irreflexive

Symmetric

Antisymmetric

Transitive